Unit 9, Lesson 1: Slides, Flips, and Spins

Benchmark

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

Learning Targets

- ☐ I can show the mapping of a preimage to an image.
- ☐ I can describe relationships between figures after a slide, flip, or spin.

9.1.1 Warm-Up: Nutritionist

Sasha Bard is a registered dietitian nutritionist who works for the U.S. Department of Agriculture. A nutritionist, or dietitian, is an expert in food, health, and nutrition. Nutritionists are health professionals, so they work in a variety of settings. They work in hospitals helping patients, in grocery stores helping customers, and in gyms to help athletes perform their best.

1. What is something that surprised you about what a nutritionist does?

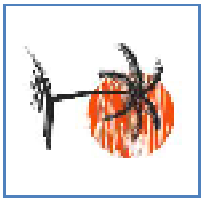
Sasha recommends that students who may be interested in pursuing this field should get out and volunteer at a food pantry or soup kitchen.

2. How do you think volunteering at a food pantry or soup kitchen would be helpful for someone interested in being a nutritionist?

9.1.2 Collaboration: Describing Movements

Felix is using palm tree wallpaper squares to create a design on his wall. He experiments with different movements by sliding, flipping, and spinning squares like the one shown.

1. For each movement shown in the table, identify the description of the movement Felix used.

	Movement		Description
A			o slide o flip o spin
B			o slide o flip o spin
C			o slide o flip o spin

2. Compare your responses with your partner and resolve any differences.

3. Choose one of the movements and describe how you determined its type.

I determined movement

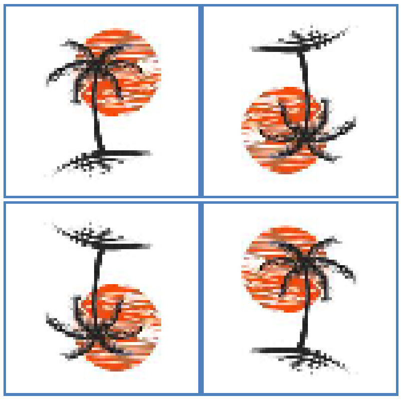
- o A
- o B
- o C

showed a

- o slide
- o flip
- o spin

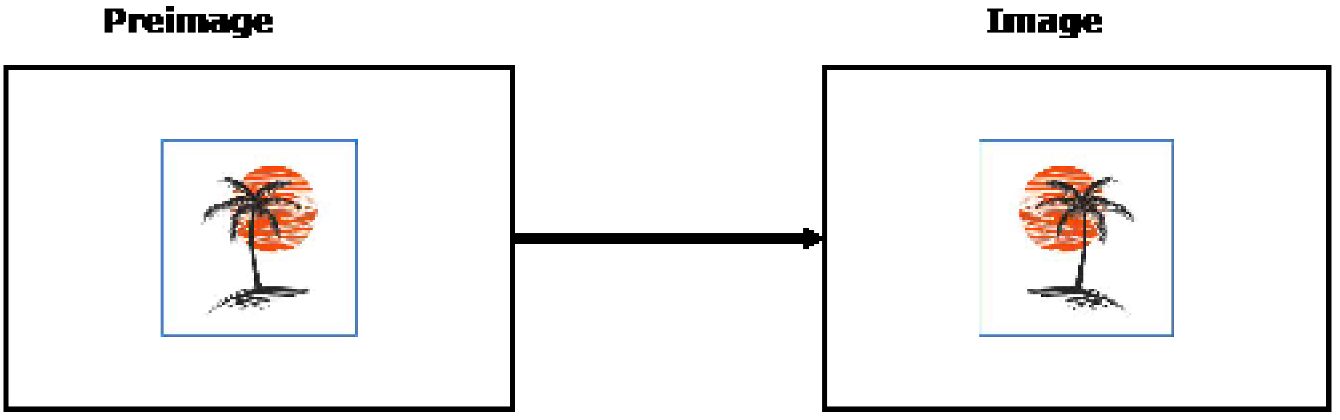
because ...

4. Describe which type(s) of movements you think Felix used to create the design shown.

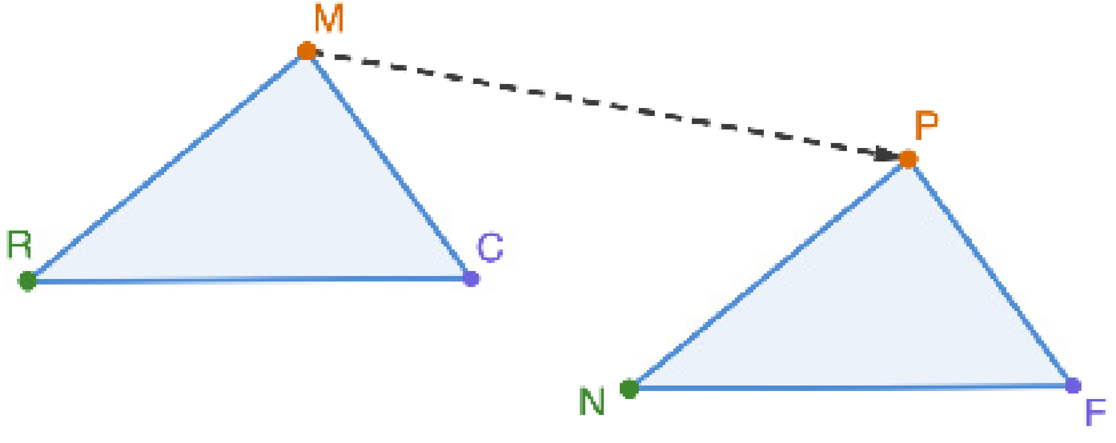


9.1.3 Guided Instruction: Preimages, Images, and Mapping

In geometry, when a figure is transformed from one position to another, the original figure is called the **preimage** and the figure that results after some kind of movement is called the **image**.



Triangle MCR is shown. Consider ΔMCR the preimage. Imagine a copy of ΔMCR slides along the dotted ray shown to create a new image, ΔPFN.



1. Discuss with your partner what you notice about the relationship between the preimage and the image. Then, write a brief summary of your discussion.

Notice that point *N* on the preimage resulted in point *P* on the image. These are called corresponding points.

2. Complete each statement.

a. Point *N* on the
- o image
- o preimage

corresponds to point *P* on the
- o image.
- o preimage.

b. Point *F* on the
- o image
- o preimage

corresponds to point
- o *R*
- o *C*

on the
- o image.
- o preimage.

c. Point *R* on the
- o image
- o preimage

corresponds to point
- o *N*
- o *F*

on the
- o image.
- o preimage.

The ray from point M to point P shows the movement of the preimage to the image. This is called a mapping and shows how one point is "mapped" onto another.

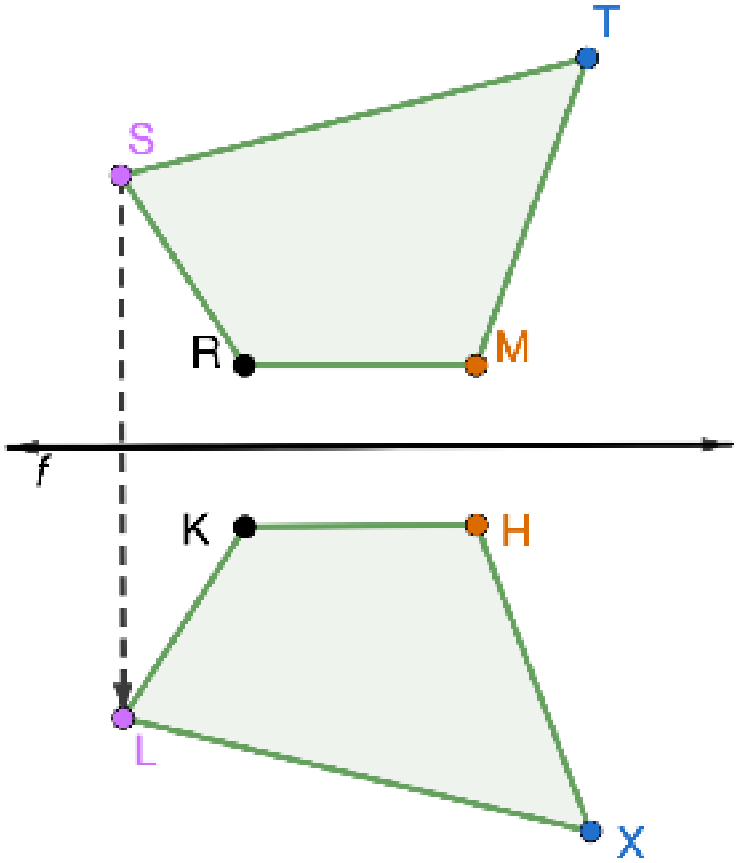
A **mapping** is a standardized form of showing how the preimage moves, or transforms, to the image.

3. Draw rays on the diagram to show the mapping of points C and R to their corresponding points in the image.

The three rays together now show the mapping of the preimage, ΔMCR to the image, ΔPFN because all of the vertices are shown to have the same movement.

9.1.4 Exploration: Images and Preimages in Slides, Flips, and Spins

1. Quadrilateral *MRST* is flipped across line *f* to create quadrilateral *HKLX* so that point *S* aligns with point *L*.



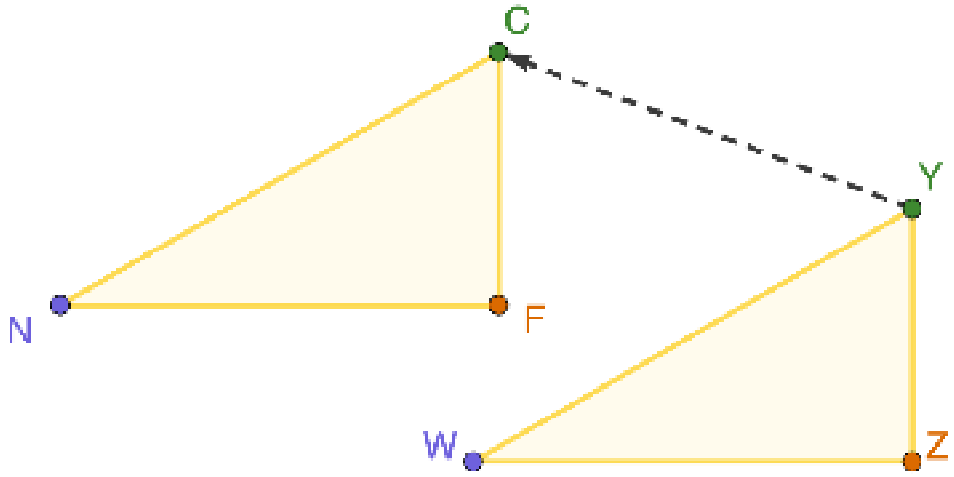
a. The dotted ray shows the mapping of point *S* to point *L*. Complete the mapping of the vertices of quadrilateral *MRST* to the vertices of quadrilateral *HKLX*.

b. **Complete the table by working with your partner to find each measure using rulers and protractors.**

Side Length	Measure (cm)	Side Length	Measure (cm)
Segment MR		Segment HK	
Segment RS		Segment KL	
Segment ST		Segment LX	
Segment TM		Segment XH	
Angle	**Measure (°)**	**Angle**	**Measure (°)**
$\angle MRS$		$\angle HKL$	
$\angle RST$		$\angle KLX$	
$\angle STM$		$\angle LXH$	
$\angle TMR$		$\angle XHK$	

c. **What do you notice about the side lengths and angle measures of quadrilateral $MRST$ and quadrilateral $HKLX$?**

2. The image, ΔNCF, is shown after sliding the preimage, ΔWYZ, along the dotted ray.



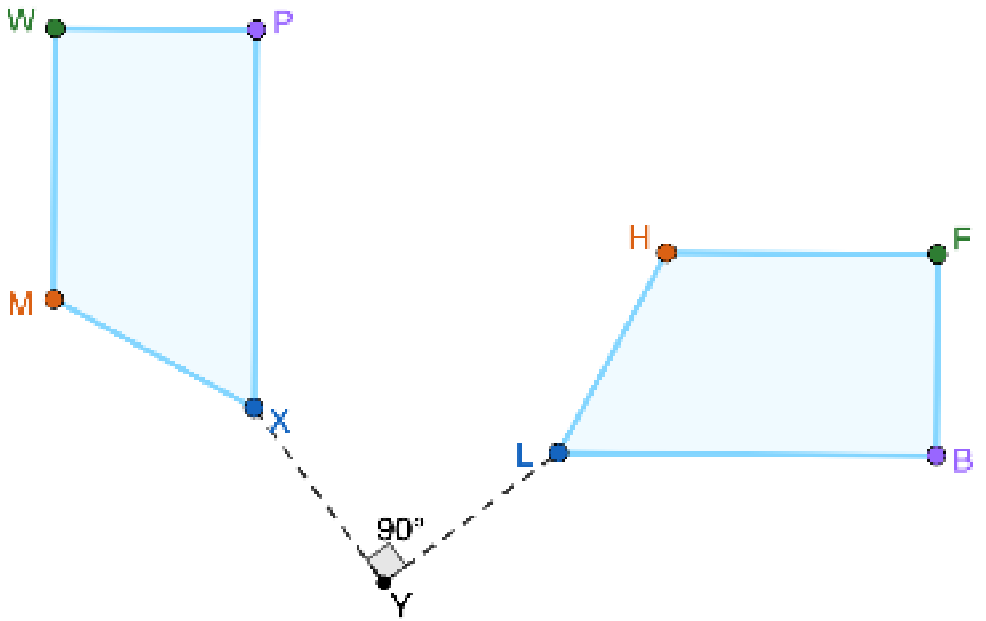
a. Complete the mapping from ΔWYZ to ΔNCF.

b. Complete the table by working with your partner to find each measure using rulers and protractors.

Side Length	Measure (cm)	Side Length	Measure (cm)
Segment WY		Segment NC	
Segment YZ		Segment CF	
Segment WZ		Segment NF	
Angle	**Measure (°)**	**Angle**	**Measure (°)**
$\angle WYZ$		$\angle NCF$	
$\angle YZW$		$\angle CFN$	
$\angle ZWY$		$\angle FNC$	

c. Discuss the relationship between the side lengths and angle measures of ΔWYZ and ΔNCF with your partner.

3. Trapezoid *LHFB* is spun around point *Y* to create trapezoid *XMWP*.



The angle shown, 90°, provides a mapping of one set of vertices for the spin. Each point on trapezoid *LHFB* forms a 90° with the corresponding point on trapezoid *XMWP* with point *Y* as its vertex.

Complete the table by working with your partner to find each measure using rulers and protractors.

Side Length	Measure (cm)	Side Length	Measure (cm)
Segment *LH*		Segment *XM*	
Segment *HF*		Segment *MW*	
Segment *FB*		Segment *WP*	
Segment *BL*		Segment *PX*	
Angle	**Measure (°)**	**Angle**	**Measure (°)**
∠*BLH*		∠*PXM*	
∠*LHF*		∠*XMW*	
∠*HFB*		∠*MWP*	
∠*FBL*		∠*WPX*	

9.1.5 Bring It Together: Describing the Relationship Between Figures After a Slide, Flip, or Spin

Review the three movements explored in today's lesson – slides, flips, and spins.

1. Identify whether each movement preserves distance (side length) and angle measure. Complete the table below by selecting "yes" or "no" based on your observations.

Movement	Is distance (side length) preserved?	Are angle measures preserved?
Slide	o Yes o No	o Yes o No
Flip	o Yes o No	o Yes o No
Spin	o Yes o No	o Yes o No

Figures are **congruent** if all corresponding parts of the figures are congruent.

2. Complete the statements.

When a slide, flip, or spin is performed on a figure, the corresponding side lengths of the preimage and image [o are / o are not] the same length and the angle measures [o are / o are not] the same measure. Therefore, the preimage and image are [o congruent / o similar] figures.

9.1.6 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and how much sense you made of today's learning targets.

- ☐ **I can show the mapping of a preimage to an image.**
- ☐ **I can describe relationships between figures after a slide, flip, or spin.**